Unit 6, Lesson 17

An Algorithm with Partial Quotients

Standards

Addressing **4.NBT.B.6**

Instructional Routines

- Number Talk

Goals

- Comprehend (in spoken language) the meaning of the term "partial quotients."
- Explain (orally and in writing) strategies for dividing three-digit numbers by one-digit numbers.
- Interpret (orally and in writing) recording methods for division that uses partial quotients.

Student Facing Learning Goals

Let's make sense of and use an algorithm that uses partial quotients.

Lesson Purpose

The purpose of this lesson is to introduce students to ways to record partial quotients when dividing multi-digit numbers.

Narrative

Previously, students have found quotients by decomposing a dividend and finding the quotient for each decomposed part until all of the dividend is divided. Students also reasoned in terms of multiplication—adding partial products until they reach the value of the dividend—and in terms of place value. They also used area diagrams and base-ten diagrams—among other representations—to support their reasoning.

In this lesson, students use partial quotients and a couple of ways to record them systematically—by writing a series of equations, and by using an algorithm that uses partial quotients.

Access for Students with Disabilities

- Action and Expression

Access for English Learners

- MLR8

Required Materials

Materials to Gather

- Base-ten blocks: Activity 1

Lesson Timeline

Warm-up	10 min
Activity 1	20 min
Activity 2	15 min

Teacher Reflection Questions

Today's lesson encouraged small-group collaboration. How did students interact with each other's ideas today in the work? Who was heard in their group? Who was not heard? How can you ensure in future small-group collaborations that all student's voices are heard?

Synthesis Estimate	10 min
Cool-down	5 min

Warm-up

10 min

Number Talk: Divide by 3

Standards

Addressing **4.NBT.B.6**

Instructional Routines

- Number Talk

This *Number Talk* encourages students to look for and make use of the structure of numbers in base-ten to mentally solve division problems (MP7). The reasoning elicited here will be helpful later in the lesson when students divide large numbers using increasingly more abstract strategies. In explaining their reasoning, students practice being precise in their use of language (MP6).

Student Task Statement

Find the value of each expression mentally.

- $90 \div 3$
- $96 \div 3$
- $960 \div 3$
- $954 \div 3$

Student Response

- 30: I know that 90 is 10×9 and $9 \div 3$ is 3, so $90 \div 3$ is $10 \times 9 \div 3$, which is 30.
- 32: Ninety is 30×3 and 6 is 2×3, so 96 is $(30 \times 3) + (2 \times 3)$, which is $(30 + 2) \times 3$ or 32×3.
- 320: I know that $96 \div 3$ is 32, so $960 \div 3$ is $10 \times 96 \div 3$ or 10×32, which is 320.
- 318: I know that 954 is $960 - 6$, so it is $(320 \times 3) - (2 \times 3)$, which is $(320 - 2) \times 3$ or 318×3.

Launch

- Groups of 2
- Display one expression.
- "Give me a signal when you have an answer and can explain how you got it."
- 1 minute: quiet think time

Activity

- Record answers and strategy.
- Keep expressions and work displayed.
- Repeat with each expression.

Activity Synthesis

- "How did each expression help you find the next one?"
- Consider asking:
  - "Who can restate _____ 's reasoning in a different way?"
  - "Did anyone have the same strategy but would explain it differently?"
  - "Did anyone approach the expression in a different way?"
  - "Does anyone want to add on to___'s strategy?"

Activity 1

20 min

Decompose Dividends

Standards

Addressing **4.NBT.B.6**

The purpose of this activity is for students to encounter a way to divide a multi-digit number by using partial quotients and writing equations for them. They analyze and interpret the equations and consider how it is like and unlike finding quotients using base-ten representations. In the next activity, students will be introduced to a way to record partial quotients vertically.

Access for English Language Learners

MLR8 Discussion Supports. Provide students with the opportunity to rehearse describing their strategy for finding $465 \div 5$ with a partner before they share with the whole class.
Advances: Speaking

Required Materials

Materials to Gather

- Base-ten blocks: Activity 1

Student Task Statement

1. Find the value of $465 \div 5$. Explain or show your reasoning. You may use base-blocks if you find them helpful.
2. Here is how Priya finds the value of $465 \div 5$.

$$\begin{aligned} 400 \div 5 &= 80 \\ 60 \div 5 &= 12 \\ 5 \div 5 &= 1 \\ \hline 465 \div 5 &= 93 \end{aligned}$$

   a. Describe the steps in Priya's method.

   b. How is Priya's method similar to your method?

   c. Use Priya's method to find the value of $428 \div 4$.

Student Response

1. 93. Sample response: There are 4 hundreds, 6 tens, and 5 ones, to be put into 5 equal groups.

   The 5 tens and 5 ones can be divided into 5 groups.

1 ten 1 one	1 ten 1 one	1 ten 1 one	1 ten 1 one	1 ten 1 one

   The 4 hundreds can be decomposed into 40 tens and

Launch

- Groups of 4
- Give students access to base-ten blocks.

Activity

- 3–5 minutes: independent work time on the first question
- 2–5 minutes: Pause after the first question and discuss students' responses. Record and display responses for all to see.
- 5–7 minutes: independent work time on remaining problems

Activity Synthesis

- Invite students to share their interpretations of Priya's work and compare it to their reasoning in the first question.
  - "How did Priya decompose the number 465?" (By place value. $400 + 60 + 5$)
  - "What does Priya do after writing the first three

then split across 5 groups, 8 tens in each.

1 ten 1 one 8 tens 2 ones	1 ten 1 one 8 tens 2 ones	1 ten 1 one 8 tens 2 ones	1 ten 1 one 8 tens 2 ones	1 ten 1 one 8 tens 2 ones

The 1 ten can be decomposed into 10 ones, which means 2 ones per group.

In each group, there are 9 tens and 3 ones.

2. Sample response:
   a. Priya divided 465 into 5 groups in smaller parts. First, she divided 400 by 5, which gave 80. Then she divided 60 by 5, which gave 12. Lastly, she divided 5 by 5, which is 1. The quotient is $80 + 12 + 1$, which is 93.
   b. They are similar because the division into equal groups is done using hundreds, tens, and ones.
   c.
$$\begin{aligned} 400 \div 4 &= 100 \\ 20 \div 4 &= 5 \\ 8 \div 4 &= 2 \\ \hline 428 \div 4 &= 107 \end{aligned}$$

equations?" (She adds up the quotients.)

- "We can find a quotient in parts—dividing a portion of the dividend at a time—until there is no more (or until there is not enough) of the dividend to divide."
- "Each quotient is called a partial quotient."

Activity 2

15 min

Tyler's Method

Standards

Addressing **4.NBT.B.6**

The purpose of this activity is to introduce students to an algorithm that uses partial quotients, a vertical method of recording partial quotients. Students compare and contrast this approach with other ways of dividing numbers using partial quotients. They also use partial quotients to divide multi-digit numbers.

When students analyze Priya and Tyler's work and explain their reasoning, they critique the reasoning of others (MP3).

Access for Students with Disabilities

Action and Expression: Internalize Executive Functions. Invite students to verbalize their strategy for problem 2 before they begin. Students can speak quietly to themselves, or share with a partner.
Supports accessibility for: Conceptual Processing, Organization

Student Task Statement

Tyler uses a different method to find the value of

Launch

- Groups of 4

$465 \div 5$. Let's compare Priya's and Tyler's work.

Priya's method

$$\begin{aligned} 400 \div 5 &= 80 \\ 60 \div 5 &= 12 \\ 5 \div 5 &= 1 \\ \hline 465 \div 5 &= 93 \end{aligned}$$

Tyler's method

$$\begin{array}{rl} \boxed{93} & \\ 1 & \\ 12 & \\ 80 & \\ 5\overline{)465} & \\ -\ 400 & 5 \times 80 \\ \hline 65 & \\ -\ 60 & 5 \times 12 \\ \hline 5 & \\ -\ 5 & 5 \times 1 \\ \hline 0 & \end{array}$$

1. How are these methods alike? How are they different? List as many similarities and differences as you can find.
2. Why do you think Tyler uses subtraction in his method?
3. Show how Tyler might record the process of finding the value of $428 \div 4$.

Student Response

1. Alike:
   - The 465 is decomposed into parts and each part is divided by 5 separately.
   - The partial quotients are 80, 12, and 1.
   - The 80, 12, and 1 are added up to get 93.
   - Both methods show 5 groups of 80, 5 groups of 12, and 5 groups of 1.

   Different:
   - Priya wrote division equations to show the divisor and partial quotients.
   - Tyler wrote multiplication expressions. He stacked the partial quotients vertically.
   - Tyler subtracted numbers several times. There is no subtraction in Priya's work.
   - Priya showed 5 groups of 80 in $400 \div 5 = 80$. Tyler showed it in 5×80.
2. To keep track of what is left after he removed a partial amount that has been divided by 5. (For example, after dividing 400 by 5, he removed it from 465 to get 65.)
3. Sample response:

- Display Priya and Tyler's methods.

Activity

- "Tyler used a different method to record $465 \div 5$. Analyze what is happening in his method. Think about how the two methods are alike and different."
- 3 minutes: independent work time on the first two questions
- 3 minutes: small-group discussion
- Invite students to share their analyses on the two methods.
- If not mentioned by students, highlight that both Priya and Tyler divided in parts, but reasoned and recorded differently. Both methods are simply two ways to record partial quotients, but they are not fundamentally different.
  - Priya recorded the partial quotients with division equations.
  - The partial quotients in Tyler's work are recorded as factors being multiplied by 5, and also listed above the dividend.
  - Tyler kept dividing in parts and subtracting until there's nothing left of the dividend to divide.
- Clarify the meaning of the numbers in Tyler's method before students work on the last question.
- 3 minutes: independent work time to find the value of $428 \div 4$.

Activity Synthesis

- Invite students to share their responses. Highlight the different ways to decompose 428 or the different partial quotients that could be used to find $428 \div 4$.
- Make sure students see that Priya's equations and Tyler's method are simply two ways to record partial quotients, but they are not fundamentally different.
- "Tyler's vertical recording method is another type of algorithm."

107			107		
2			7		
5			50		
100			50		
4)428			4)428		
− 400	4×100		− 200	4×50	
28			228		
− 20	4×5		− 200	4×50	
8			28		
− 8	4×2		− 28	4×7	
0			0		

Lesson Synthesis

"Today we learned to use an algorithm that uses partial quotients to divide numbers."

"How would you explain 'partial quotients' to a classmate who might be absent today?" (We can find a quotient in parts—dividing a portion of the dividend at a time—until there is no more or until there is not enough of the dividend to divide. Each quotient is called a partial quotient.)

"Suppose we'd like to find the value of $738 \div 9$ and know we could decompose the 738 into parts. How would we know what numbers to choose?" (Look for multiples of 9. Try to start with the largest multiple of 9 and 10 within 738.)

"What are some ways to decompose 738 into multiples of 9?" ($720 + 18$, or $450 + 270 + 18$, among others.)

Display:

$$738 \div 9$$

$$\begin{aligned} 720 \div 9 &= 80 \\ 18 \div 9 &= \ \ 2 \\ \hline 738 \div 9 &= 82 \end{aligned}$$

82		
2		
80		
9)738		
− 720	9×80	
18		
− 18	9×2	
0		

"We saw two ways of recording partial quotients—by writing a series of equations and by recording the steps of division vertically. Where can we see the partial quotients in each one?"

Suggested Centers

- Compare (1–5), Stage 4: Divide within 100 with One-Digit Divisors (Supporting)
- Rolling for Fractions (3–5), Stage 2: Multiply a Fraction by a Whole Number (Supporting)

Cool-down

5 min

Subtract Groups

Standards

Addressing **4.NBT.B.6**

Student Task Statement

What is the value of $430 \div 5$? Explain or show your reasoning.

Student Response

86. Sample responses:

400 is 80 groups of 5 and 30 is 6 groups of 5. Adding the groups of 5—80 and 6—gives the quotient.

$$\begin{aligned} 400 \div 5 &= 80 \\ 30 \div 5 &= 6 \\ \hline 430 \div 5 &= 86 \end{aligned}$$

$$\begin{array}{rl} \boxed{86} & \\ 6 & \\ 20 & \\ 60 & \\ 5\overline{)430} & \\ -\ 300 & 5 \times 60 \\ \hline 130 & \\ -\ 100 & 5 \times 20 \\ \hline 30 & \\ -\ 30 & 5 \times 6 \\ \hline 0 & \end{array}$$

Responding to Student Thinking

Students find correct partial quotients but don't find their sum, or don't connect the sum to the value of $430 \div 5$.

Next Day Supports
Launch the *Warm-up* or first activity by highlighting important notation from previous lessons.